

Project Initialization and Planning Phase

Date	27 June 2026
Team ID	
Project Title	EduGenie – AI Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

EduGenie is an AI-powered learning platform designed to enhance the educational experience by providing intelligent assistance to students. The system integrates Google's Gemini API with FastAPI services to deliver accurate explanations, summaries, quizzes, and personalized learning roadmaps.

Project Overview	
Objective	Develop an intelligent educational assistant capable of generating study materials, answering questions, explaining concepts, summarizing content, and creating personalized learning pathways.
Scope	The project focuses on: • Question Answering • AI Summarization • Quiz Generation • Concept Explanation • Personalized Roadmaps

	• Learning History • Analytics Dashboard • Student Progress Tracking
Problem Statement	Students often struggle to manage educational resources efficiently due to information overload, lack of personalized learning support, and absence of adaptive educational systems.
Impact	EduGenie improves: • Learning efficiency • Knowledge retention • Revision speed • Student engagement • Self-paced learning • Academic performance
Proposed Solution	
Approach	The platform processes educational queries through Gemini AI and FastAPI services. Responses are analyzed and presented using an interactive dashboard. User history is maintained for personalized recommendations.
Key Features	✓ AI Question Answering ✓ Concept Explainer ✓ Quiz Generator ✓ Content Summarizer ✓ Roadmap Builder ✓ Dashboard Analytics ✓ History Tracking ✓ Authentication ✓ Personalized Recommendations □

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	NVIDIA T4 GPU (Kaggle), 2 vCPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	20 GB Available Storage
Software		
Frameworks	Python frameworks	Streamlit, TensorFlow
Libraries	Additional libraries	Transformers, TensorFlow, Pandas, NumPy, NLTK, Scikit-learn, Plotly
Development Environment	IDE	Visual Studio Code, Jupyter Notebook, Kaggle
Data		

Data	Source, size, format	Kaggle Emotion Dataset, CSV format
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